PlantCare AI

Advanced Plant Disease Detection Using Deep Learning (MobileNetV2)

Live Demo: https://omkumar04-plant-care-ai.hf.space/

GitHub Repository: https://github.com/Om-Pramod-Kumar/Plant-Care-AI

1. Introduction

PlantCare AI is a deep learning powered web application designed to detect plant diseases from leaf images. The system allows users to upload a plant leaf image and receive instant predictions about possible diseases affecting the plant. The application uses transfer learning with MobileNetV2 to classify plant diseases across multiple categories.

Early detection of plant diseases is essential to prevent crop loss and improve agricultural productivity. PlantCare AI provides an easy-to-use interface for farmers, gardeners, and agricultural professionals to quickly diagnose plant health using artificial intelligence.

2. Problem Statement

Plant diseases are one of the leading causes of crop loss worldwide. Identifying plant diseases manually requires expert knowledge and time. Many farmers do not have access to agricultural experts for timely diagnosis. An automated AI-based solution can help detect diseases quickly and accurately using plant leaf images.

3. Objectives

- Detect plant diseases using deep learning
- Use transfer learning with MobileNetV2 for accurate predictions
- Provide an easy web interface for image upload
- Assist farmers in early disease detection
- Reduce crop loss through early diagnosis

4. System Architecture

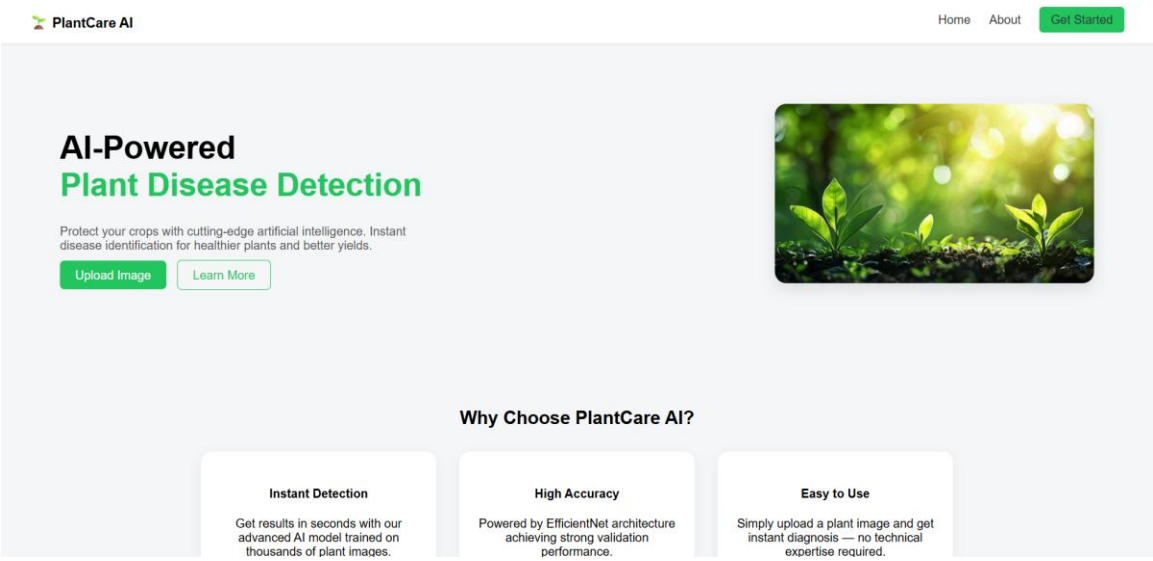
The system follows a client-server architecture. Users upload plant leaf images through the web interface. The frontend sends the image to the Flask backend API. The backend preprocesses the image and passes it to the trained MobileNetV2 model for prediction. The

prediction result and confidence score are returned to the frontend and displayed to the user.

Flow: User → Frontend → Flask Backend → MobileNetV2 Model → Prediction Result

5. User Interface

Home Page Interface



About Technology Section

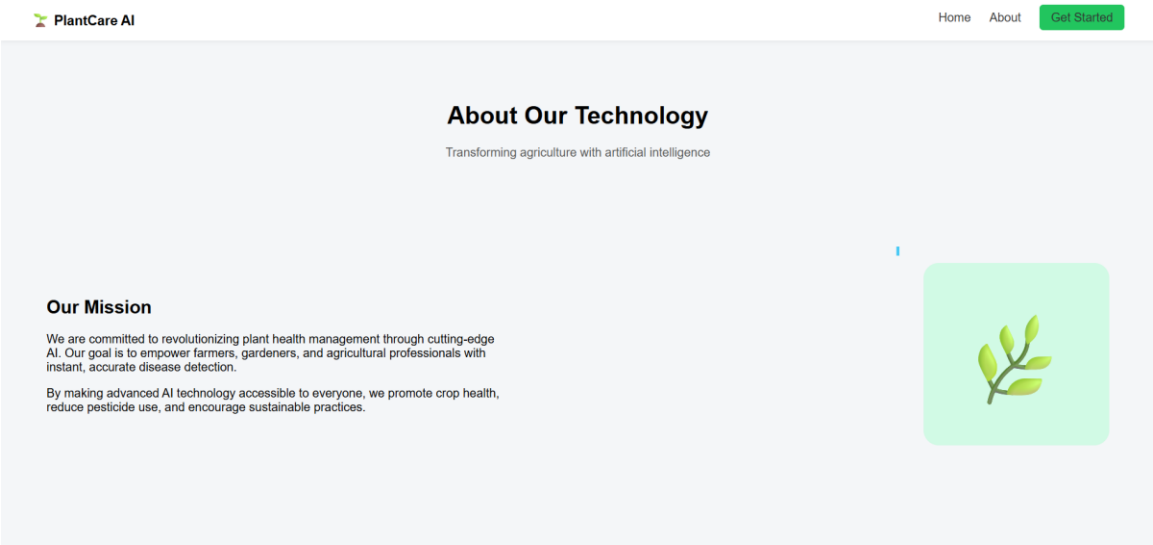
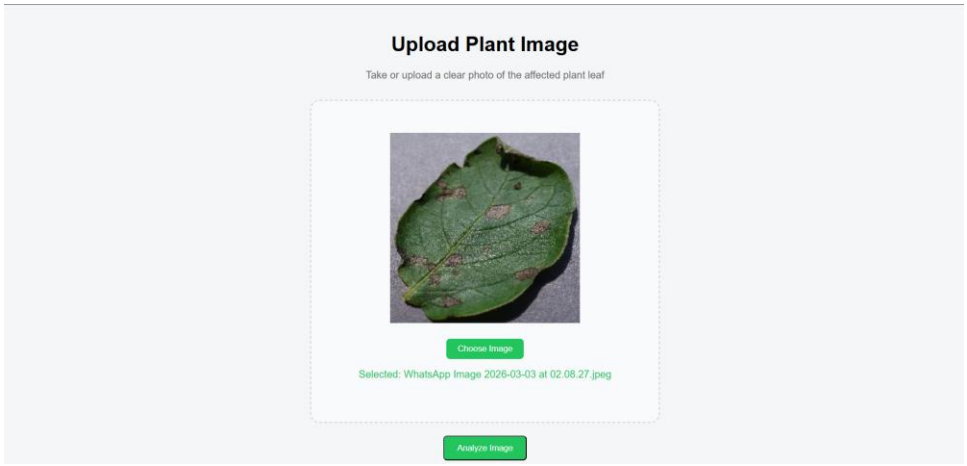
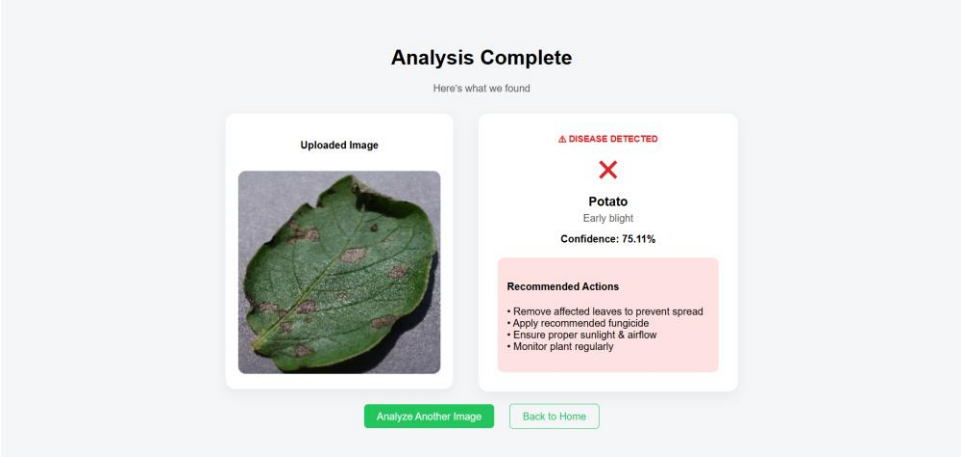


Image Upload Interface



Prediction Result Interface



6. Backend API

The backend of the application is built using Flask. It handles image uploads, model inference, preprocessing, and returning prediction results.

Main API Endpoints:

/ → Home Page

/about → About Page

/upload → Upload Interface

/predict → Image Prediction API (POST)

/result → Display Prediction Result

7. Image Processing Pipeline

1. User uploads plant leaf image

2. Image converted to RGB format

3. Image resized for model input

4. Pixel normalization using MobileNet preprocessing

5. Image converted to NumPy array

6. Passed to trained MobileNetV2 model

7. Prediction and confidence score returned

8. Deep Learning Model

The system uses MobileNetV2 architecture with transfer learning. MobileNetV2 is a lightweight convolutional neural network designed for efficient image classification. It provides a good balance between accuracy and performance making it suitable for real-time web applications.

Model Architecture:

Input Layer (224x224x3) → MobileNetV2 Base Model → GlobalAveragePooling2D → Dense Layer → Softmax Output (38 Classes)

9. Dataset

The model was trained using the New Plant Diseases Dataset from Kaggle. The dataset contains more than 87,000 labeled plant leaf images across 38 plant disease categories including healthy leaves.

10. Technology Stack

Frontend: HTML5, CSS3, JavaScript

Backend: Flask

Machine Learning: TensorFlow, Keras

Deep Learning Model: MobileNetV2

11. Key Features

• AI powered plant disease detection

- Upload plant leaf images
- Instant disease prediction
- Confidence score display
- Recommended treatment actions

12. Future Improvements

- Mobile application version
- IoT based plant monitoring
- More plant disease classes
- Admin analytics dashboard

13. Conclusion

PlantCare AI demonstrates how deep learning and computer vision can help solve real-world agricultural problems. By leveraging MobileNetV2 and transfer learning, the system provides fast and accurate plant disease detection that can assist farmers in improving crop health.